Periodic Pattern Learning in the Surrogate Network

Spectral Bias, Fourier Features, and Input Encoding for the CPG-Driven Cosserat Rod

Snake-HRL Project — Reference Document

1 The Problem

The neural surrogate f_θ replaces the transition operator $T\colon \mathbb{R}^{129} \to \mathbb{R}^{124}$ that composes 500 Position Verlet substeps of a Cosserat rod simulation. During those substeps, the CPG (Central Pattern Generator) produces a *time-varying* rest curvature at each Voronoi node:

$$\kappa_v^{\text{rest}}(t^n) = A \sin\!\big(2\pi k \cdot s_v + 2\pi f \cdot t^n + \phi_0\big) + b, \qquad v = 1, \ldots, 19 \tag{1}$$

where $(A, f, k, \phi_0, b) \in \mathbb{R}^5$ is the RL action, held fixed for the entire control interval $\Delta t_{\text{ctrl}} = 0.5\,\text{s}$, and $t^n = t_{\text{step}} + n\,\Delta t$ advances at every substep. Because of the $2\pi f \cdot t^n$ term, the 500 substeps are *not identical*: each sees a different curvature configuration, and the rod responds dynamically to the propagating wave.

The surrogate must therefore learn the *integrated effect* of this oscillatory forcing:

$$\mathbf{s}_{t+1} = T(\mathbf{s}_t, \mathbf{a}_t) = \underbrace{V_{\Delta t}\big(\,\cdot\,;\ \boldsymbol{\kappa}^{\text{rest},\,499}\big) \ \circ\ \cdots\ \circ\ V_{\Delta t}\big(\,\cdot\,;\ \boldsymbol{\kappa}^{\text{rest},\,0}\big)}_{\text{500 substeps, each with distinct } \boldsymbol{\kappa}^{\text{rest}}} \big(\mathbf{s}_t\big) \tag{2}$$

This is not a power iteration $V_{\Delta t}^{500}$. The qualitative behaviour of T depends strongly on how many oscillation cycles fit within Δt_{ctrl}:

$$n_{\text{cycles}} = f \cdot \Delta t_{\text{ctrl}} \tag{3}$$

f (Hz)	n_{cycles}	Dynamics character
0.5	0.25	Slow sweep — rod barely completes a quarter-wave
1.0	0.50	Half-cycle — rod swings in one direction
2.0	1.00	Full cycle — rod nearly returns to its initial configuration
3.0	1.50	One-and-a-half cycles — rod reverses mid-step

Table 1: Qualitative effect of the CPG frequency on one RL step. At integer n_{cycles}, the angular velocity ω_z nearly returns to its initial value; at half-integer n_{cycles}, ω_z reverses sign.

The mapping $f \mapsto \Delta\omega_z$ is therefore *itself oscillatory*: the change in angular velocity is approximately zero when n_{cycles} is an integer and maximal when it is a half-integer. An MLP with standard activations must approximate this sinusoidal relationship from raw scalar inputs — precisely the setting where *spectral bias* becomes problematic.

2 Spectral Bias

Spectral bias (the "frequency principle") is the well-documented tendency of neural networks to learn low-frequency components of a target function before high-frequency components [1]. Networks with standard activations (ReLU, SiLU, tanh) preferentially fit smooth, slowly-varying functions. High-frequency patterns in the target require more training time, more parameters, or may never be fully learned.

The root cause is that the Neural Tangent Kernel (NTK) of standard MLPs has a rapid spectral falloff: the kernel's eigenvalues decay with increasing frequency, so gradient descent converges slowly on high-frequency modes.

Relevance to the surrogate. The surrogate's target function contains an oscillatory dependence on the frequency parameter f. Concretely, $\Delta\omega_z$ as a function of f is approximately sinusoidal (Table 1). An MLP receiving f as a raw normalized scalar in $[-1, 1]$ must learn this periodicity purely from data. This is consistent with the poor angular velocity prediction observed empirically ($R^2 = 0.23$ on ω_z components).

Severity assessment. The spectral bias risk is **localised** to the frequency and wavenumber dimensions of the action space. The spatial CPG phase is already handled by the per-element encoding (Section 3), which provides explicit sinusoidal features. The remaining gap is the *frequency-domain* relationship between (f, k) and the integrated dynamical response.

3 The Existing Per-Element Phase Encoding

The current input to the surrogate is a 193-dimensional vector:

$$\mathbf{x} = \big[\, \underbrace{\mathbf{s}_{\text{rel}}}_{\mathbb{R}^{128}} \;\big|\; \underbrace{\mathbf{a}}_{\mathbb{R}^{5}} \;\big|\; \underbrace{\gamma^{\text{phase}}}_{\mathbb{R}^{60}} \,\big] \in \mathbb{R}^{193} \tag{4}$$

where $\mathbf{s}_{\text{rel}}$ is the relative rod state (CoM-subtracted positions, heading as $\sin / \cos$, velocities, yaw, ω_z), $\mathbf{a}$ is the raw action, and γ^{phase} contains 60 per-element CPG phase features:

$$\gamma^{\text{phase}} = \big[\, \underbrace{\sin(\psi_j)}_{j=1}^{20} \;\big|\; \underbrace{\cos(\psi_j)}_{j=1}^{20} \;\big|\; \underbrace{A\sin(\psi_j)}_{j=1}^{20} \,\big] \in \mathbb{R}^{60} \tag{5}$$

with $\psi_j = 2\pi k \cdot s_j + 2\pi f \cdot t_{\text{step}} + \phi_0$ being the CPG phase angle at element j at the *start* of the RL step.

3.1 Strengths

(a) **Spatially resolved.** Each element gets its own phase, capturing the traveling wave's spatial structure. The network can distinguish head-to-tail curvature patterns rather than seeing a single global phase.

(b) $(\sin, \cos)$ **pair avoids phase wrapping.** The two-component representation is continuous everywhere, avoiding the 2π discontinuity that would arise from providing the raw angle ψ_j.

(c) **Commanded curvature is explicit.** The $A \sin(\psi_j)$ features directly provide the rest curvature κ_j^{rest} at step start — the actual forcing term that enters the constitutive relation $\boldsymbol{\tau}_v^L = \mathbf{B}_{\text{blk}} (\boldsymbol{\kappa}_v - \boldsymbol{\kappa}_v^{\text{rest}})$.

(d) **Structurally equivalent to Fourier features.** The encoding is a hand-crafted Fourier feature expansion of the CPG phase at each spatial location, with exactly the right frequency (the CPG fundamental). This directly addresses spectral bias for the spatial-temporal phase.

3.2 Limitations

(a) **Fundamental frequency only.** The encoding provides $\sin(\psi_j)$ and $\cos(\psi_j)$ at the CPG fundamental. The *integrated response* of 500 nonlinear substeps generates harmonics — the rod's actual curvature at step end is not a pure sinusoid even when the forcing is. Cross-frequency terms $\sin(2\psi_j)$ and $\sin(\psi_j)\sin(\psi_k)$ are absent.

(b) **No explicit n_{cycles} representation.** The number of oscillation cycles per step (Eq. 3) is the critical quantity that determines whether ω_z increases, decreases, or reverses. It is available implicitly from f in the action, but not encoded as a periodic feature.

(c) **Phase at step start only.** The encoding provides the CPG state at t_{step}, but the integrated response depends on the phase *trajectory* over 500 substeps. The phase sweeps from $\psi_j(0)$ to $\psi_j(0) + 2\pi f \cdot \Delta t_{\text{ctrl}}$. This sweep information must be inferred from the raw scalar f.

Verdict. The existing encoding is well-designed but has a specific gap: the network must learn the oscillatory relationship between f and the integrated dynamical response from a raw scalar. Fourier-encoding f (and k) directly targets this gap.

4 Approaches Considered and Not Recommended

4.1 SIREN (Periodic Activation Functions)

SIREN [3] replaces all activation functions with $\sin(\omega_0 \cdot x)$, where ω_0 is a frequency hyperparameter (typically 30). Its derivatives are also sinusoidal, making it effective for learning signals whose derivatives must be accurate (e.g., signed distance functions, PDE solutions).

Why not here. The periodicity in the surrogate's problem is in the *input encoding*, not in the *target function*. The mapping from (body shape + CPG phase features) to (body shape delta) is not itself periodic — it is a complex nonlinear regression where the periodic structure has already been factored out via the phase encoding. SIREN also produces ringing artifacts when the target contains non-periodic components (position drift, friction effects), and requires a specialised initialisation scheme that may interact poorly with the existing LayerNorm + residual block architecture.

4.2 Snake Activation

The Snake activation [4] $\mathrm{Snake}_a(x) = x + \frac{\sin^2(ax)}{a}$ combines ReLU-like monotonicity with a periodic inductive bias. Its primary advantage is *extrapolation*: given periodic training data, Snake networks extend the pattern beyond the training domain.

Why not here. The surrogate operates within a bounded action space and state distribution. Extrapolation is not the primary concern — the body-frame representation and dense training coverage address generalisation more fundamentally. Switching activations introduces risk (different initialisation, learning rate sensitivity) with minimal expected gain since the periodicity is already in the input.

4.3 Neural ODEs

Neural ODEs [5] learn continuous-time dynamics $d\mathbf{x}/dt = f_\theta(\mathbf{x}, t)$ and integrate via an ODE solver. They are appropriate when the time horizon varies or adaptive step sizes are needed.

Why not here. The surrogate is a *fixed-step, fixed-horizon* map ($\mathbb{R}^{193} \to \mathbb{R}^{128}$). The entire point is to replace 500 integration steps with a single forward pass. A Neural ODE would reintroduce stepwise integration in the forward pass, defeating the purpose.

4.4 Fourier Neural Operators (FNO)

FNO [6] learns mappings between function spaces via spectral convolutions on spatial grids, excelling at parametric PDEs with resolution invariance.

Why not here. The surrogate maps between *finite-dimensional vectors*, not functions. The 20-element rod is already discretised at a fixed resolution. FNO is architecturally mismatched — it would add complexity without benefit for this specific problem.

5 Recommended Solutions

The recommendations target the specific gap identified in Section 3.2: the raw scalar representation of the frequency f and wavenumber k in the action.

5.1 Solution 1: Random Fourier Features on (f, k)

Theory. Tancik et al. [2] showed that mapping low-dimensional inputs through sinusoidal functions before feeding them to an MLP eliminates spectral bias. The encoding is:

$$\gamma(\mathbf{v}) = \left[\, \sin\!\big(2\pi \mathbf{B}\,\mathbf{v}\big),\ \cos\!\big(2\pi \mathbf{B}\,\mathbf{v}\big) \,\right] \ \in \mathbb{R}^{2m} \tag{6}$$

where $\mathbf{v} \in \mathbb{R}^d$ is the input, $\mathbf{B} \in \mathbb{R}^{m \times d}$ is a random frequency matrix with $B_{ij} \sim \mathcal{N}(0, \sigma^2)$, and σ controls the bandwidth (the range of frequencies represented). Each row of $\mathbf{B}$ defines a random frequency direction in input space; the $(\sin, \cos)$ pair at that frequency provides the network with a basis function at the corresponding scale.

The connection to the NTK is direct: Fourier features reshape the NTK's spectrum to have uniform eigenvalues across a wider frequency band, removing the low-frequency preference.

Application. Apply Eq. 6 to the frequency and wavenumber components of the action:

$$\mathbf{v} = \big(f_{\text{norm}},\ k_{\text{norm}}\big) \in [-1, 1]^2 \qquad \gamma(\mathbf{v}) \in \mathbb{R}^{2m} \tag{7}$$

with $m = 32$ (yielding 64 additional features) and $\mathbf{B} \in \mathbb{R}^{32 \times 2}$. The bandwidth σ is a hyperparameter to be swept.

Bandwidth selection. The target function's characteristic frequency in normalised input space determines the optimal σ. The mapping $f \mapsto \Delta\omega_z$ oscillates on the scale of $1/\Delta t_{\text{ctrl}} = 2\,\text{Hz}$ in physical units, corresponding to a normalised frequency of $\mathcal{O}(1\text{–}10)$ in $[-1, 1]$ space. A sweep over $\sigma \in \{1, 3, 10, 30\}$ covers the plausible range:

σ	Effect
1	Low-frequency features only — may underfit the oscillatory dependence
3	Moderate — captures the primary oscillation
10	High-frequency — resolves fine structure in $f \mapsto \Delta\omega_z$
30	Very high-frequency — risk of overfitting to noise

Table 2: Fourier feature bandwidth σ sweep. Expected optimum: $\sigma \in [3, 10]$.

Implementation. A `FourierFeatures` module samples $\mathbf{B}$ once at initialisation and registers it as a non-trainable buffer (or optionally as a learnable parameter):

$$\boxed{\begin{aligned} &\mathbf{B} \sim \mathcal{N}(\mathbf{0},\ \sigma^2\mathbf{I}), \quad \mathbf{B} \in \mathbb{R}^{m\times 2} && \text{[sample once at init]} \\ &\mathbf{p} = 2\pi\,\mathbf{B}\,\mathbf{v} && \text{[projection, } \mathbf{p} \in \mathbb{R}^m\text{]} \\ &\gamma(\mathbf{v}) = \big[\sin(\mathbf{p}),\ \cos(\mathbf{p})\big] && \text{[output, } \gamma \in \mathbb{R}^{2m}\text{]} \end{aligned}} \tag{8}$$

The extended surrogate input becomes:

$$\mathbf{x}' = \big[\ \mathbf{s}_{\text{rel}} \ \big|\ \mathbf{a} \ \big|\ \gamma^{\text{phase}} \ \big|\ \gamma(f_{\text{norm}}, k_{\text{norm}})\ \big] \ \in \mathbb{R}^{193+2m} \tag{9}$$

For $m = 32$: $\dim(\mathbf{x}') = 257$.

Why this is the highest-impact change. The Fourier features directly provide the sinusoidal basis functions that the MLP would otherwise need to synthesise from the raw scalar f. Instead of learning $\Delta\omega_z \approx g(\sin(c \cdot f))$ for some unknown constant c, the network receives $\sin(2\pi b_i \cdot f)$ at multiple frequencies b_i and need only learn a *linear combination* of these basis functions. This converts a hard nonlinear approximation problem into a substantially easier regression.

5.2 Solution 2: Explicit n_{cycles} Feature

Add a derived feature that directly encodes how many oscillation cycles the CPG completes within one RL step:

$$n_{\text{cycles}} = f \cdot \Delta t_{\text{ctrl}} \qquad \gamma_{\text{cyc}} = \big[\sin(2\pi\, n_{\text{cycles}}),\ \cos(2\pi\, n_{\text{cycles}})\big] \ \in \mathbb{R}^2 \tag{10}$$

where f is the denormalised frequency in Hz. When n_{cycles} is an integer (full cycles), $\sin(2\pi n_{\text{cycles}}) = 0$ and $\cos(2\pi n_{\text{cycles}}) = 1$; when it is a half-integer (reversal), $\sin = 0$ and $\cos = -1$. This directly tells the network whether the step ends mid-oscillation.

The cost is 2 additional input dimensions. The frequency must be denormalised from the action before computing n_{cycles}:

$$f = f_{\min} + \frac{f_{\text{norm}} + 1}{2}\,(f_{\max} - f_{\min}) = 0.5 + \frac{f_{\text{norm}} + 1}{2} \cdot 2.5 \quad [\text{Hz}] \tag{11}$$

This is the simplest change with a clear physical motivation.

5.3 Solution 3: Harmonic Phase Features

The nonlinear rod dynamics generate harmonics of the CPG fundamental. Even though the forcing is sinusoidal at frequency k, the rod's curvature response contains $2k$, $3k$, and higher harmonics due to:

- Geometric nonlinearity (finite rotations couple curvature modes)
- Quadratic velocity terms in the damping forces
- Contact and friction interactions

The current encoding provides only the fundamental. Extending to H harmonics:

$$\gamma^{\text{harm}} = \big[\, \underbrace{\sin(h\,\psi_j),\ \cos(h\,\psi_j)}_{\text{for each } h=1,\ldots,H} \;\big|\; A\sin(\psi_j) \,\big]_{j=1}^{20} \;\in \mathbb{R}^{20(2H+1)} \tag{12}$$

For $H = 3$ (fundamental + 2nd + 3rd harmonic): $\dim(\gamma^{\text{harm}}) = 20 \times 7 = 140$, adding 80 features beyond the current 60.

The h-th harmonic provides a basis function that oscillates h times as fast along the rod body and in time, letting the network represent the nonlinear response without synthesising these harmonics internally.

6 Combined Input Architecture

Combining all three solutions, the enhanced surrogate input is:

$$\mathbf{x}^{\star} = \big[\, \underbrace{\mathbf{s}_{\text{rel}}}_{\mathbb{R}^{128}} \;\big|\; \underbrace{\mathbf{a}}_{\mathbb{R}^{5}} \;\big|\; \underbrace{\gamma^{\text{harm}}}_{\mathbb{R}^{140}} \;\big|\; \underbrace{\gamma(f,k)}_{\mathbb{R}^{64}} \;\big|\; \underbrace{\gamma_{\text{cyc}}}_{\mathbb{R}^{2}} \,\big] \;\in \mathbb{R}^{339} \tag{13}$$

Compared to the current 193-dimensional input, this adds 146 features. The MLP hidden dimension (512) and depth (3 layers) remain unchanged; only the input projection layer widens.

Component	What it encodes	Current	Proposed
Relative state $\mathbf{s}_{\text{rel}}$	Rod shape, velocities, heading	128	128
Action $\mathbf{a}$	Raw CPG parameters (normalised)	5	5
Per-element phase	CPG phase at each element (fundamental)	60	—
Harmonic phase γ^{harm}	CPG phase at each element ($H = 3$ harmonics)	—	140
Fourier features $\gamma(f, k)$	Frequency-domain encoding of f and k	—	64
Cycle encoding γ_{cyc}	$\sin$ / $\cos$ of oscillation cycles per step	—	2
Total		**193**	**339**

Table 3: Surrogate input dimensions: current vs. proposed. The harmonic phase features ($H = 3$) subsume the current fundamental-only encoding ($H = 1$).

Parameter count impact. The first hidden layer grows from $193 \times 512 = 98{,}816$ to $339 \times 512 = 173{,}568$ parameters (a 76% increase in the first layer). Total model parameters increase from ∼790k to ∼860k — a modest 9% increase. All subsequent layers are unchanged.

7 Experiment Design

The three solutions should be tested *independently* before combining, to isolate the contribution of each.

Priority	Experiment	Extra dims	Expected gain	Risk
1	n_{cycles} encoding (Eq. 10)	+2	Medium	None
2	Fourier features on (f, k), σ sweep (Eq. 7)	+64	High	Low
3	Harmonic phase features, $H = 3$ (Eq. 12)	+80	Medium	Low
4	Combined: all three	+146	High	Low

Table 4: Experiment priority. Primary metric: validation R^2 on ω_z components (current baseline: 0.23). Secondary metric: validation MSE on all 124 state dimensions.

Fourier feature σ sweep. For experiment 2, sweep $\sigma \in \{1, 3, 10, 30\}$ with $m = 32$ fixed. All other hyperparameters (learning rate, batch size, epochs) remain at their current values to isolate the effect of the encoding.

Ablation. If the combined model (experiment 4) improves R^2, ablate by removing one component at a time to confirm each contributes independently.

8 Why Not Fourier-Encode Everything?

A natural question: if Fourier features help for (f, k), why not apply them to all 128 state dimensions?

(1) **Curse of dimensionality.** Random Fourier features scale as $\mathcal{O}(2m \cdot d)$ in output dimension. For $d = 128$ and $m = 32$, this would add $128 \times 64 = 8{,}192$ features — an input dimension explosion that dominates the model.

(2) **The state is not periodic.** Spectral bias is a problem when the *target function* oscillates as a function of the *input coordinates*. The state vector (positions, velocities, yaw angles) does not enter the target as periodic arguments. The target's dependence on the state is smooth and well-captured by standard activations.

(3) **The bottleneck is localised.** The poor ω_z prediction is specifically attributable to the frequency-dependent oscillatory dynamics (Section 1). Fourier features on (f, k) target this bottleneck precisely. Applying them elsewhere dilutes the signal.

9 Summary

The CPG's time-varying rest curvature creates a frequency-domain challenge for the surrogate: the integrated effect of 500 substeps is an oscillatory function of the action parameters (f, k), and standard MLPs learn such functions slowly (spectral bias).

The existing per-element phase encoding already addresses the *spatial-temporal phase* at each element — it is a well-designed Fourier feature expansion of the CPG state. But it does not address the *frequency-domain* relationship between (f, k) and the integrated dynamical response.

Three targeted solutions are proposed:

1. **Random Fourier Features on** (f, k) (Section 5.1): Provides sinusoidal basis functions of the frequency parameters, converting a hard nonlinear approximation problem into an easier regression. Highest expected impact.

2. **Explicit** n_{cycles} **encoding** (Section 5.2): Two features that directly encode how many oscillation cycles fit in the control interval. Trivial to implement, physically motivated.

3. **Harmonic phase features** (Section 5.3): Higher harmonics of the per-element phase, capturing the rod's nonlinear response to sinusoidal forcing.

SIREN, Snake activations, Neural ODEs, and Fourier Neural Operators were considered and rejected: the periodicity is in the *input encoding*, not the *network architecture*, and the surrogate's fixed-step, finite-dimensional structure does not match these approaches' strengths.

References

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